

Proposal quality and review choices

Known contract gaps now have concrete proposed answers.

Position and model attitude are source inputs. Detailed transform, dataset, instance and scene-chart choices still need author review.

64 regressions and 61 distinguishing controls passed.
213,231 coordinate results per Python/C++ path.

No new computed source properties. Group adoption and whole-proposal conformance are not claimed.

Authored inputs and computed results

Authored inputs	Meaning in the proposed model
crs:position	Absolute 3D model origin in source WKT components, units and height reference.
crs:orientation	Physical geodetic ENU model attitude; identity fallback. Distinct from projected grid convergence.
Ordinary xformOps	Intentional scale, pivots and project adjustment. Descendant stacks retain model-local meaning.
Native data association	Asset, format, field and coordinate domain. Values, masks and observation times remain native.
Computed results	Resolution, convergence, projection scale and bounds are results; no crs:scale source property.

This is a complete review candidate. Tests do not establish author agreement to the detailed choices.

Anchor adjustments and model-local descendants

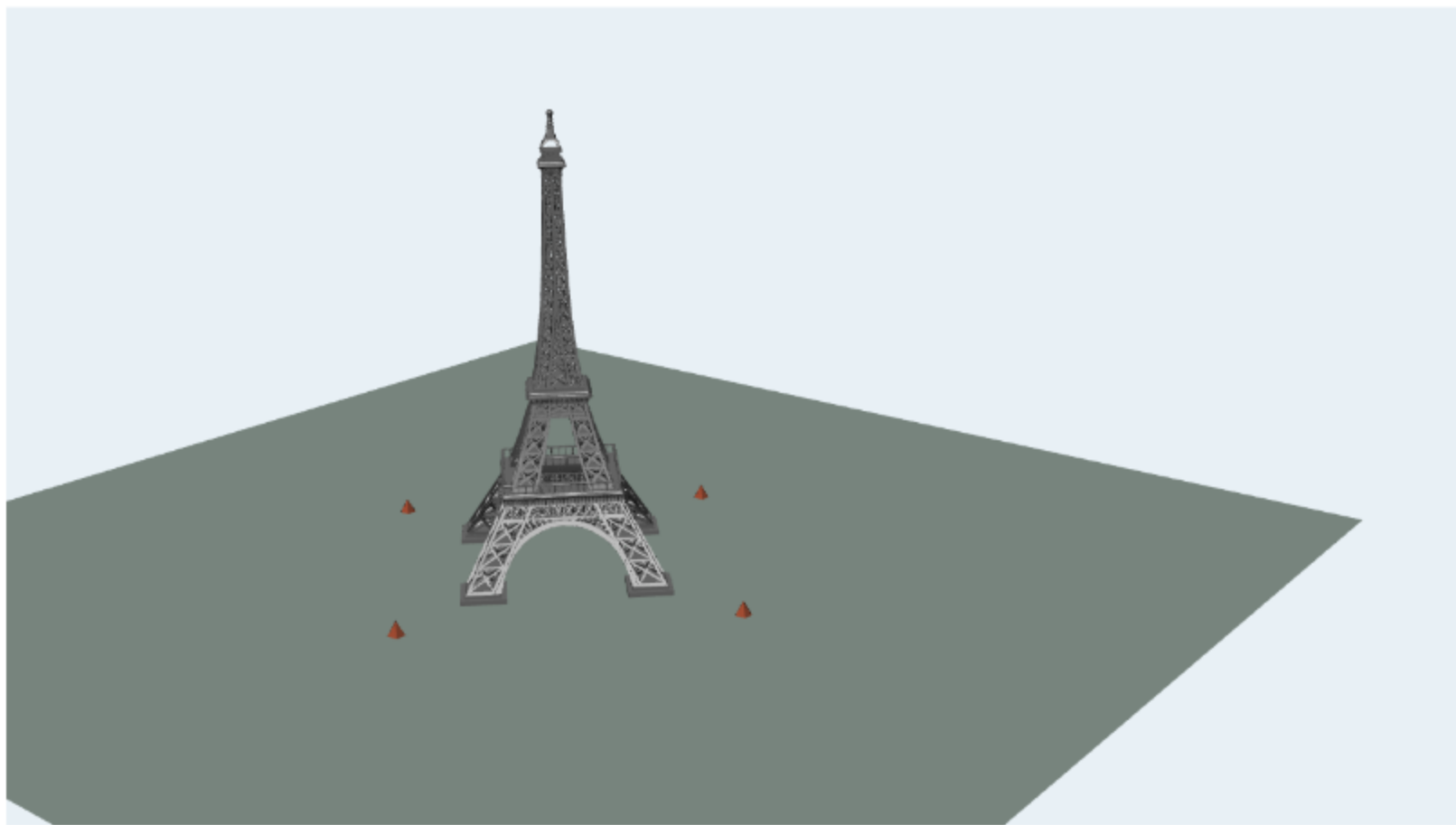
A geographic asset sits inside a projected site. Its model orientation turns local X northward.

Queried position	Easting (m)	Northing (m)	Height (m)
Anchor with +10 site adjustment	448262.0014	5411954.9099	35.0000
Child with +3 local X	448262.0292	5411957.9087	35.0000
Child with ordinary reset	448252.0292	5411957.9087	35.0000

The +10 follows site easting. The child offset follows the oriented model.
The reset removes the ordinary adjustment and retains CRS placement.

The raw working-axis child alternative differs by 4.222 m. Descendant semantics are proposed for review.

Eiffel geometry in native Hydra / Storm



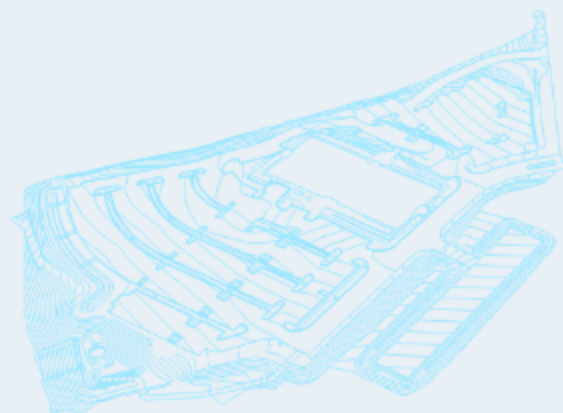
163,440 vertices

C++ pointwise results
163,416 shading normals
Fresh USD export readers
Native Storm color output

Illustrative plane and markers.
No surveyed Paris context.

The renderer consumes an ordinary resolved export. This run does not install a live geospatial scene-index filter.

Colorado survey in a calibrated site



14,359 vertices

Original LandXML
breaklines

WKT-carried site
calibration

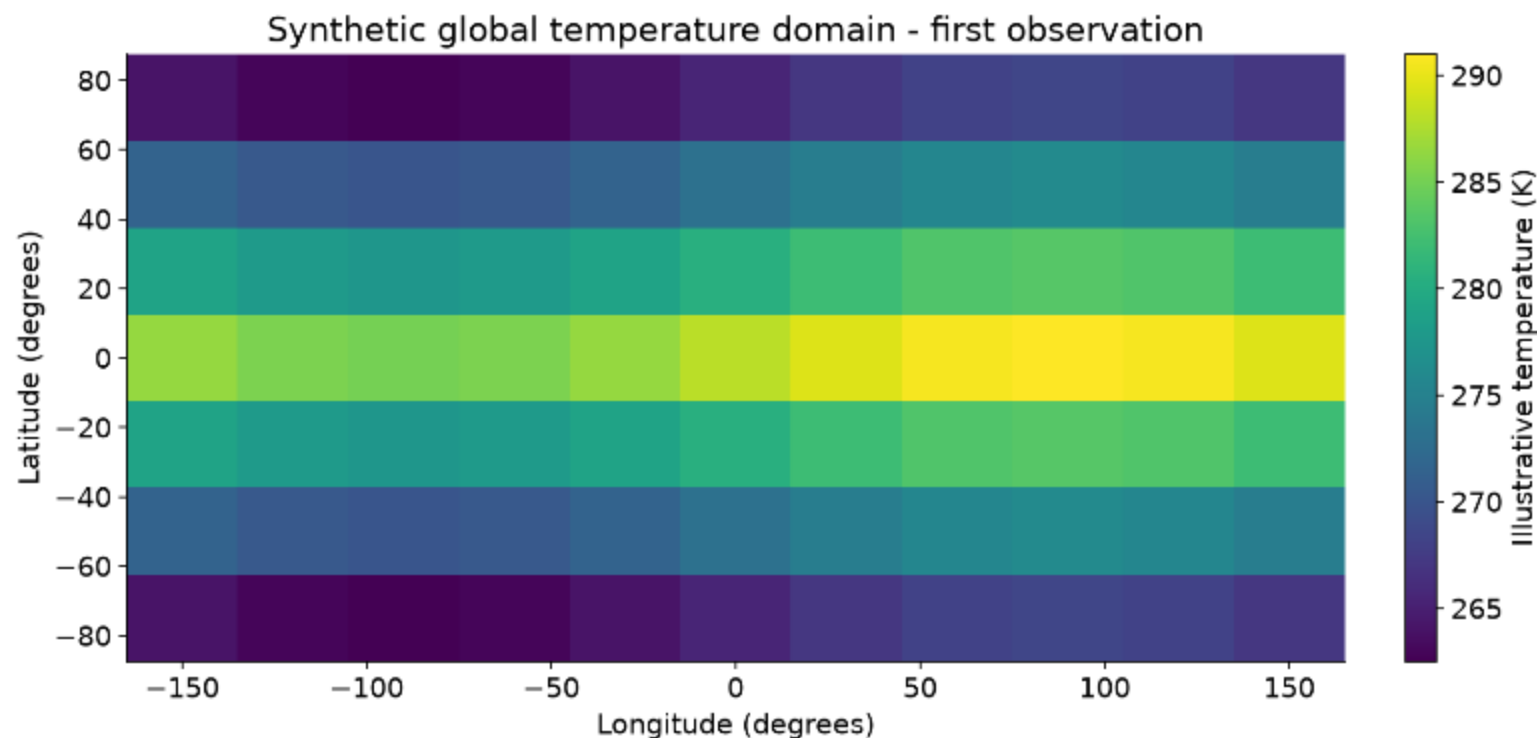
Declared US survey feet

Separate stage
conventions

No terrain surface
was fabricated.

Fresh readers verify the resolved geometry. Polygonal bounds cover returned vertices and straight faces, with no continuous-surface certificate.

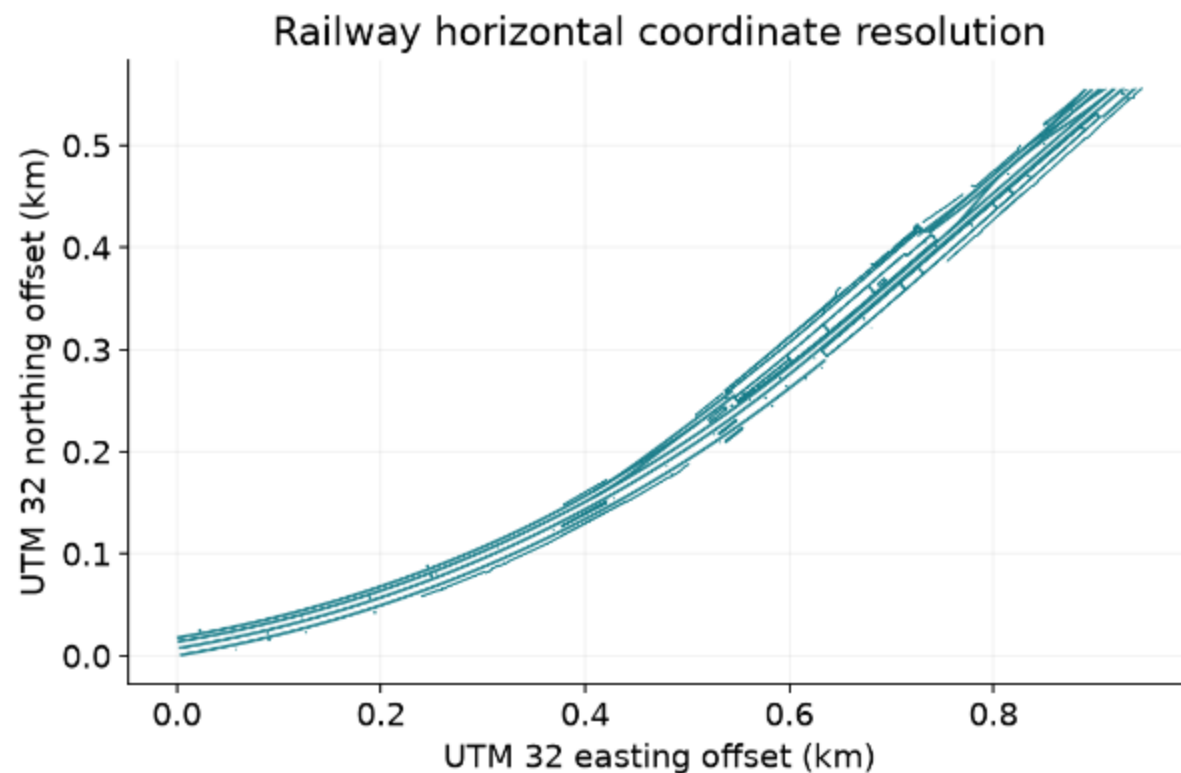
Global measurements in geographic and ECEF coordinates



154 synthetic 3D samples retain values and two observation times.

Explicit 100 m ellipsoidal height; illustrative temperature in kelvin. Both readers resolve geographic and ECEF outputs. Observation time is distinct from epoch.

Railway geometry retains its source coordinates



15,822 vertices

2,386 railway parts
Explicit horizontal source copy
UTM 32 output coordinates
Original data retained

Original third components
remain uninterpreted.

The larger-scale workflows keep useful 2D resolution without inventing height. Plot offsets serve display only.

The same partner data in both placements

France: CC49 to Lambert-93 (metres)

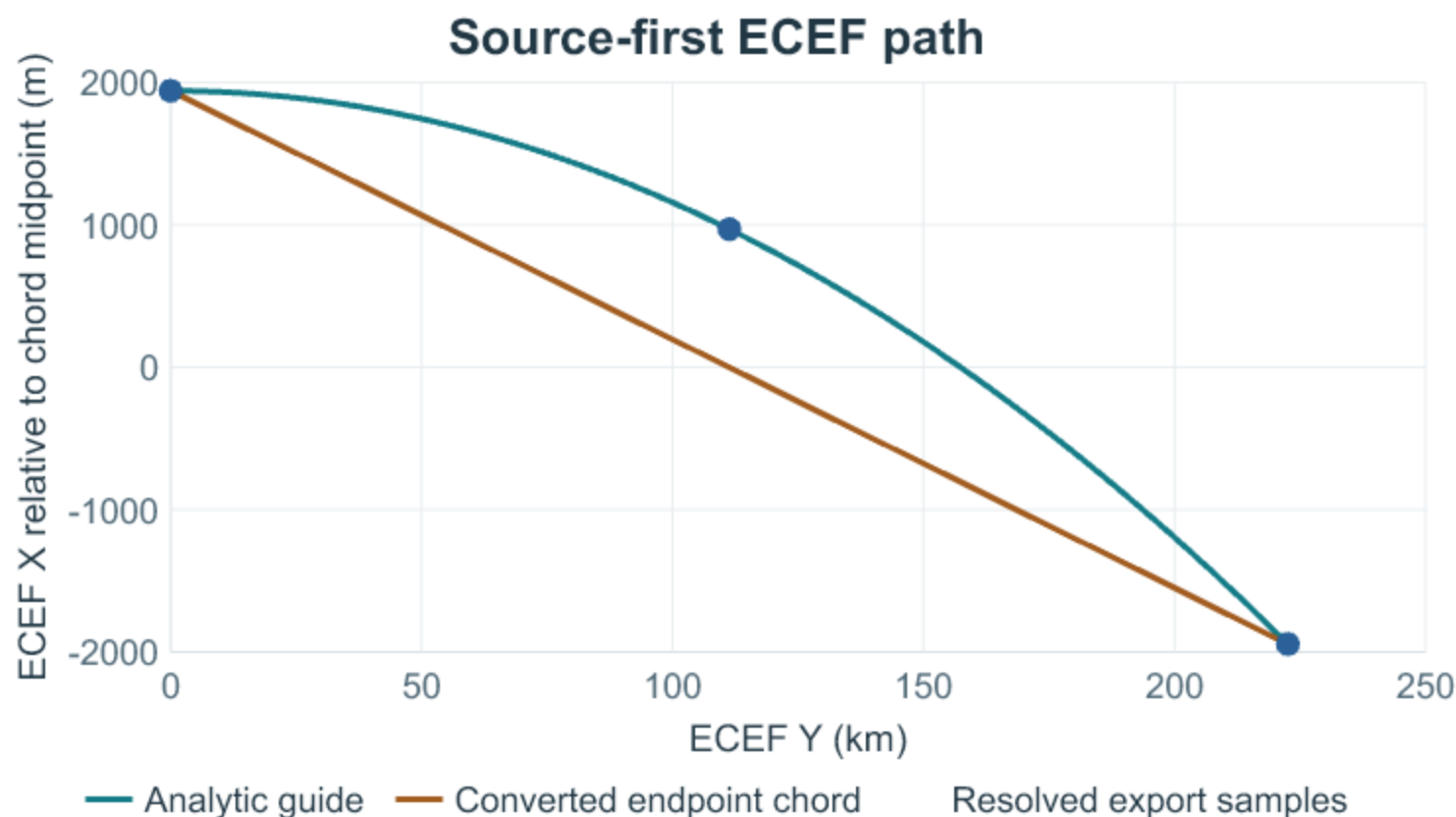
Component	Source	USD Python	USD C++	OV-hosted Python
Easting	1661099.0390	661101.1094	661101.1094	661101.1094
Northing	8180053.1170	6857834.2565	6857834.2565	6857834.2565
Height	37.2980	37.2980	37.2980	37.2980

Colorado: State Plane to calibrated site (US survey feet)

Component	Source	USD Python	USD C++	OV-hosted Python
Easting	3108002.1122	3108871.4501	3108871.4501	3108871.4501
Northing	1752265.0666	1206317.4694	1206317.4694	1206317.4694
Height	5482.9424	5484.6779	5484.6779	5484.6779

Both USD implementations agree. OV reuses Python placement. All paths share PROJ; these comparisons do not certify survey accuracy.

Source interpolation and sampled exports



971.421 m

Midpoint difference from interpolating only the converted endpoints.

Fresh geometry exports
Sample keys 0, 5, 10
timeCodesPerSecond = 48

Both readers verify each exported sample. No original-trajectory guarantee is claimed between samples. Chart axes use different distance scales.

Verified consumers and remaining evidence limits

Consumer	Evidence reaching its output
USD Python and C++	Independent placement arithmetic; full point maps, Cartesian frame estimates and polygonal bounds. Shared PROJ and external decoding.
Omniverse live geometry	177,864 vertices read back from runtime buffers/matrices. 36 edit, failure and recovery checks. Reuses Python placement.
Hydra / Storm	Two ordinary resolved exports reach points/transform data sources and the native renderer. No live CRS filter.

All 11 exports pass fresh-reader checks. Geographic scene charts and bound prototypes are covered.

Remaining evidence limits: continuous nonlinear certificates, general primvar coverage and physics integration. Detailed candidate choices still need author review.